**How do I use if statements and loops in a Drafter website?**

You can use if statements and for loops inside your Drafter route functions just like you would in regular Python! They are especially useful for your concrete calculator because you can check calculations, display messages, and show all your saved concrete pads.

Let's look at some examples.

**1. Using if statements in Drafter**

An if statement lets your website make decisions based on information.

For example, let's say you want your concrete calculator to tell someone whether they have enough concrete.

```python
@route
def check_concrete(state: State) -> Page:
    if state.cubic_yards >= 5:
        message = "You need at least 5 cubic yards!"
    else:
        message = "You need less than 5 cubic yards."

    return Page(state, [
        Header("Concrete Estimate"),
        message,
        Button("Go Home", "index")
    ])
```

**What's happening?**

Python checks whether state.cubic_yards is greater than or equal to 5.

If it is, the first message is displayed.

Otherwise, the second message is displayed.

The message is added to the webpage using Page().

**2. Using if statements to check checkboxes**

Remember how we discussed checkboxes?

You can use if statements to add extra costs or materials.

For example:

```python
@route
def calculate(state: State, reinforcement: bool) -> Page:
    total = state.concrete_cost

    if reinforcement:
        total += 150

    return Page(state, [
        Header("Your Estimate"),
        "Total cost: $" + str(total)
    ])
```

If the reinforcement checkbox is checked, $150 is added. If it isn't checked, nothing is added. The result is displayed on the website.

**3. Using for loops in Drafter**

A for loop lets you repeat something multiple times.

For your concrete calculator, this is useful for displaying all your saved concrete pads.

```python
@route
def saved_pads(state: State) -> Page:
    content = [
        Header("Saved Concrete Pads")
    ]

    for pad in state.saved_pads:
        content.append(Header(pad.name, 2))
        content.append("Length: " + str(pad.length))
        content.append("Width: " + str(pad.width))

    return Page(state, content)
```

**What's happening?**

Imagine your saved pads list contains three concrete pads: Driveway, Patio, and Garage.

The loop repeats for each pad:

- First loop: Displays Driveway.
- Second loop: Displays Patio.
- Third loop: Displays Garage.

You don't need to write separate code for every concrete pad!

**4. Combining if statements and for loops**

This is where things get really useful.

Let's say you want your saved concrete pads page to display whether each pad needs a large or small amount of concrete.

```python
@route
def saved_pads(state: State) -> Page:
    content = [
        Header("Saved Concrete Pads")
    ]

    for pad in state.saved_pads:
        cubic_feet = pad.length * pad.width * pad.thickness
        cubic_yards = cubic_feet / 27

        content.append(Header(pad.name, 2))
        content.append("Concrete needed: " + str(round(cubic_yards,
2)) + " cubic yards")

        if cubic_yards >= 5:
            content.append("Large concrete project")
        else:
            content.append("Small concrete project")

    return Page(state, content)
```

Now your website:

- Loops through every saved concrete pad.
- Calculates the cubic yards for each one.
- Checks whether that pad needs 5 or more cubic yards.
- Displays a different message depending on the result.

**5. What if your list is empty?**

You can also use an if statement to check whether someone has saved any concrete pads.

```python
@route
def saved_pads(state: State) -> Page:
    content = [
        Header("Saved Concrete Pads")
    ]

    if not state.saved_pads:
        content.append("You haven't saved any concrete pads yet.")

    else:
        for pad in state.saved_pads:
            content.append(Header(pad.name, 2))
            content.append(str(pad.length) + " feet long")

    return Page(state, content)
```

This is a really useful pattern!

`if not state.saved_pads` checks whether the list is empty.

If it is empty, the website displays a message.

Otherwise, the loop displays all the saved pads.

**6. Quick practice: What will this code display?**

Look at this Python code:

```python
yards = 6

if yards >= 5:
    message = "Large Project"
else:
    message = "Small Project"
```

- Small Project
- Large Project
- Both messages

**Remember these patterns**

| Python | What it does |
|---|---|
| `if` | Checks a condition |
| `else` | Runs when the condition is false |
| `elif` | Checks another condition |
| `for` | Repeats code for each item |
| `if not my_list` | Checks if a list is empty |
| `content.append()` | Adds something to your webpage |

My recommendation: Practice using if statements and for loops together in your saved concrete pads page. It's a great way to learn how Python logic controls what appears on your Drafter website!

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**How do I make a button that takes you to another page in Drafter?**

In Drafter, you use Button() to create a button that takes someone to another page.

The button connects to another function using that function's name.

**1. Create your home page**

```python
@route
def index(state: State) -> Page:
    return Page(state, [
        Header("Home Page"),
        "Welcome to my website!",
        Button("View Saved Pads", "saved_pads")
    ])
```

"View Saved Pads" is the text displayed on the button.

"saved_pads" is the name of the page you want to visit.

**2. Create the page the button leads to**

You need another route with the matching name.

```python
@route
def saved_pads(state: State) -> Page:
    return Page(state, [
        Header("Saved Concrete Pads"),
        "Here are your saved concrete pads!",
        Button("Go Back Home", "index")
    ])
```

`saved_pads` is the name of your second page.

The Go Back Home button takes you back to `index`.

**3. How the pages connect**

Home Page

Welcome to my website!

Try clicking the buttons to see how navigation works.

**4. The most important thing to remember**

The format is:

```python
Button("Button Text", "route_name")
```

For example:

```python
Button("Calculate Concrete", "calculate")
Button("View Saved Pads", "saved_pads")
Button("Settings", "settings")
```

Each button needs a matching `@route` function.

Tip: You can put buttons on any page, so your users can move around your website without having to return to the home page every time!